

SYNOPSIS

ON

SportsSync Venue Connect

IN PARTIAL FULFILLMENT OF

MASTER OF COMPUTER APPLICATION

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1. INTRODUCTION

In today's busy world, finding and booking sports turfs can be a big challenge for both players and turf owners. That's why we created **SportsSync Venue Connect** an easy-to-use web app that makes booking turfs and managing them much simpler. **SportsSync Venue Connect** is not just about booking grounds. It helps players find available turfs, book them quickly, and view their schedules, all in one place. Turf owners can manage their bookings, track how often their turfs are used, and make sure everything runs smoothly.

Booking turfs, checking availability, and keeping track of everything can feel overwhelming. But **SportsSync Venue Connect** makes it easy. The app keeps everything organized, gives real-time updates, and shows important details to help turf owners and admins make better decisions. Our goal is to create a platform that not only saves time but also connects players, owners, and admins in a way that makes sports more enjoyable for everyone. Whether you're booking a turf, managing it, or overseeing the platform, **SportsSync Venue Connect** has all the tools you need to stay organized and keep things running smoothly.

2. Existing System and Need for System

2.1.Existing System

2.1.1. Manual Booking: Players have to call, visit turfs, or send messages to check availability and book. This takes a lot of time and is not always reliable.

2.1.2. No Centralized System: Turf owners handle bookings and schedules manually. This can lead to confusion, errors, or even double bookings.

2.1.3. Payment Issues: Payments are often done offline, making it hard to track or manage them properly.

2.1.4. Lack of Records: There is no proper way to keep track of turf usage, bookings, or payments for future reference.

2.1.5. Admin Challenges: Admins find it difficult to manage multiple turfs, monitor owners, and ensure smooth communication between players and owners.

2.1.6. Inefficiency: The whole process is slow, unorganized, and limits how well players, owners, and admins can work together.

2.2.Need for System

2.2.1. Simplify Turf Booking: Players need an easy way to find,

check, and book available turfs without wasting time.

2.2.2. Centralized Management: Turf owners need a single platform to manage bookings, schedules, and payments in an organized way.

2.2.3. Avoid Errors: A system is needed to prevent double bookings, missed payments, and other manual errors.

2.2.4. Track Usage: Owners and admins need proper records of turf usage, bookings, and payments to make better decisions.

2.2.5. Save Time: Automating the booking and management process will save time for players, owners, and admins.

2.2.6. Better Communication: A platform is needed to connect players, owners, and admins easily and efficiently.

2.2.7. Enhance User Experience: The system should provide a smooth and reliable experience for everyone involved.

3. Scope and Objectives of System

3.1. Turf Booking: The system allows players to view available turfs and book them easily through an online platform.

3.2.Management for Turf Owners: Turf owners can manage their turfs, bookings, and payments through a single system.

3.3.Admin Control: Admins can oversee all turf bookings, monitor usage, and manage both players and owners.

3.4.Real-time Updates: Players and owners get real-time updates on bookings, availability, and changes.

3.5.Reports and Insights: The system generates reports on bookings, turf usage, and payments, helping owners and admins make informed decisions.

3.6. Multiple Roles: The system supports different user roles, including Admin, Owner, and Player, each with their own set of features.

4. Objectives of SportsSync Venue Connect:

4.1.Simplify Turf Booking: Make it easier for players to find and book turfs without any hassle.

4.2.Efficient Management: Help turf owners manage their bookings, payments, and schedules in an organized way.

4.3.Prevent Errors: Avoid issues like double bookings and missed payments by automating processes.

4.4.Improve Communication: Ensure smooth communication between players, owners, and admins through the system.

4.5.Increase Turf Utilization: Help owners track turf usage and optimize bookings to get the most out of their resources.

4.6.Enhance User Experience: Provide a simple, fast, and enjoyable experience for all users.

5. List of Modules/ Functionalities with description:

1. User Module

- **What it does:** This module allows players to sign up, log in, and book turfs easily.
- **Features:**
 - View available turfs.
 - Book turfs based on availability.
 - See past and upcoming bookings.
 - Get confirmation and reminders for bookings.

2. Owner Module

- **What it does:** This module helps turf owners manage their turfs and bookings.
- **Features:**
 - Add and update turf details (location, pricing, etc.).
 - View and manage player bookings.
 - Track payments and generate reports.
 - Set turf availability and rules.

3. Admin Module

- **What it does:** This module lets admins manage everything on the platform.
- **Features:**
 - Manage user and owner accounts.
 - View and control all bookings.
 - Generate reports about bookings and payments.
 - Handle issues or complaints from users.

4. Booking and Payment System

- **What it does:** This system handles bookings and payments.
- **Features:**
 - Secure online payment for bookings.
 - Create booking invoices.
 - Update bookings in real-time.
 - Support for different payment methods (cards, wallets, etc.).

5. Notification System

- **What it does:** This system sends notifications to users about their bookings.
- **Features:**
 - Send booking confirmations and cancellations.
 - Send reminders for upcoming bookings.
 - Notify about changes in turf availability.

6. Reporting System

- **What it does:** This system helps owners and admins generate reports.
- **Features:**
 - Booking and payment history reports.
 - Turf usage and occupancy reports.
 - Revenue and financial reports.

7. User Management

- **What it does:** This module helps admins manage user accounts.
- **Features:**
 - Add or delete user accounts.
 - Assign roles (admin, owner, player).
 - Handle account verification and password recovery.

6. System requirement specification (SRS) (Operating Environment Software and Hardware)

6.1.Operating Environment

Environment	Details
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Server Operating System	Linux (e.g., Ubuntu, CentOS) or Windows Server Recommended version: Latest stable release
Client Operating System	Windows, macOS, or Linux Recommended version: Any version that supports modern browsers
Network Requirements	Stable internet connection for cloud-based features (if applicable) Local network connection for on-premises deployment

6.2. Software Requirements

Software	Details
Operating System	Windows 7 or above / macOS / Linux
Web Server	Apache HTTP Server or Nginx (Latest stable version)
Frontend	React, Tailwind CSS, DaisyUI, Redux
Backend	Node.js, Express.js
Database Management	MongoDB
Payments	Razorpay
Image Hosting	Cloudinary
Code Editor/IDE	Visual Studio Code, Sublime Text, or similar IDE
Version Control	Git (with GitHub, GitLab, or Bitbucket for repository)
Browser Compatibility	Google Chrome, Mozilla Firefox, Safari (latest versions)

6.3. Hardware Requirements

Hardware Component	Minimum Requirements	Recommended Requirements
Server Hardware	• CPU: Dual-core processor • RAM: 4 GB • Storage: 50 GB HDD/SSD	• CPU: Quad-core processor or better • RAM: 8 GB or more • Storage: 100 GB SSD or more
Client Hardware	• CPU: Dual-core processor • RAM: 2 GB • Storage: 10 GB HDD/SSD • Network Interface: Ethernet or Wi-Fi connection	• CPU: Quad-core processor or better • RAM: 4 GB or more • Storage: 20 GB SSD or more • Network Interface: Ethernet or Wi-Fi connection

7. Proposed System for SportsSync Venue Connect:

The **SportsSync Venue Connect** system will be an online platform for booking and managing turfs, with three main types of users: **Admin**, **Owner**, and **Player**.

1. Easy to Use:

- The system will be simple and easy for everyone to use.
- Players can easily find and book turfs, and owners can manage their turfs and bookings.
- Admins will be able to manage everything on the platform.

2. Booking and Payment:

- Players will be able to book turfs based on availability and pay securely through Razorpay.
- Owners will get real-time updates on bookings and payments.
- Invoices will be automatically created for bookings, and payments will be tracked.

3. Different Roles for Users:

- Admins will have full control over everything, like managing users, turfs, and bookings.
- Owners can manage their own turfs and bookings but won't see admin-only features.

- Players can only book turfs and see their own booking history.

4. Notifications:

- The system will send notifications about bookings, cancellations, and reminders to players and owners.
- Players and owners will receive updates by email and SMS.

5. Reports and Data:

- The system will generate reports about bookings, turf usage, and payments to help owners and admins.
- Admins will have full access to all reports and data to manage the platform better.

6. Able to Grow:

- The system will be designed to grow with more users and turfs over time.
- Cloud-based services will help the system handle more traffic without problems.

7. Safe and Reliable:

- The system will use strong security features, including encryption for payments.
- Backups will be taken regularly to protect data and ensure the system is always available.

8. Feasibility Study for SportsSync Venue Connect

1. Technical Feasibility

- **Technology Stack:** Frontend (React, Tailwind CSS, DaisyUI), Backend (Node.js, Express.js), Database (MongoDB), Payments (Razorpay), Image Hosting (Cloudinary).
- **Skills Needed:** Proficiency in JavaScript, Node.js, MongoDB, payment integration, and web security.
- **Challenges:** Integrating Razorpay, Cloudinary, and real-time data syncing between frontend and backend.
- **Security:** Ensure encryption, secure payment handling, and GDPR compliance.

2. Economic Feasibility

- **Costs:** Initial costs for software, development, and cloud services. Long-term savings from automating turf bookings and payments.
- **Revenue Streams:** Turf listing fees, transaction fees, premium owner features.
- **ROI:** Short-term from increased bookings, long-term profitability through platform scalability.

3. Legal Feasibility

- **Data Protection:** Ensure **GDPR**(General Data Protection

Regulation) compliance and proper data consent.

- **Security:** Use encryption for sensitive data and comply with **PCI DSS (Payment Card Industry Data Security Standard)** for payments.
- **Compliance:** Follows local laws regarding online payments, membership management, and data protection.

4. Operational Feasibility

- **Ease of Use:** User-friendly for all roles (Admins, Owners, Users).
- **Training:** Minimal training required for all users due to intuitive design.
- **Integration:** Smooth transition from manual processes to the platform, streamlining turf management.